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QatarEnergy

QatarEnergy– IS INDUSTRIAL HYGIENE PROTECTION: PROCEDURE FOR CONTROL OF HAZARDOUS SUBSTANCES

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1.0 PURPOSE

The purpose of this procedure is to outline a process to protect the health & hygiene of personnel by anticipating, recognizing, evaluating, and controlling workplace environmental factors (chemical/physical/ biological agents) which may cause illness, impaired health & hygiene and/or significant discomfort and inefficiency among workers on QatarEnergy-IS locations.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to all QatarEnergy-IS locations. QatarEnergy-IS is focused on providing a safe workplace for all personnel including employees, contractors and visitors.

In assessing exposures to chemical/physical/biological agents for this Industrial Hygiene Procedure Program (IHPP), QatarEnergy-IS has adopted the current Threshold Limit Values (TLV's) recommended by the American Conference of Governmental Industrial Hygienists (ACGIH), or in some cases, other internationally recognized Standards.

3.0 ROLES AND RESPONSIBILITIES

HSE MANAGER

- Monitor implementation and ensure compliance to this procedure

OCCUPATIONAL HEALTH & PARAMEDIC (OHP)

- Oversee the implementation of the required routine surveillance programs for those workers identified as potentially being exposed to excessive levels of chemical, physical and biological agents as determined by methods in this procedure or relevant procedures
- Advise QatarEnergy-IS on issues dealing with Industrial Hygiene

INDUSTRIAL HYGIENIST (IH)

- Conduct industrial hygiene surveys of QatarEnergy-IS facilities every 3 years.

MANAGERS/SUPERVISORS

- Direct personnel to inform HSE Department and Head of Warehouse and Materials of any new chemical or trade-name chemical products that will be used in activities.
- Follow MOC Procedures to ensure that proper reviews are conducted and that MSDS have been delivered to and reviewed by QatarEnergy-IS.
- Forward copies of purchase orders for any radioactive materials or ionizing radiation sources to the QatarEnergy-IS Head, Environment & Regulatory Affairs and the QatarEnergy-IS QA/QC section.

4.0 PROCEDURAL STEPS

4.1 HAZARD ANTICIPATION

The practice of anticipating that a process, task, use of a chemical/product, or the operation of a specific piece of equipment will present a potential for an exposure, greater than QatarEnergy limits, to a chemical/physical/biological agent, with a consequent adverse health & hygiene effect, shall be incorporated into the Industrial Hygiene Protection Program (IHPP). The effort shall be a part of all safety reviews and included in the Management of Change (MOC) process for controlling process, safety, environmental, and workplace hygiene problems.

To ensure that the principal of hazard anticipation is practiced, the MOC process shall be utilized for any changes in process chemicals or the use and handling of items posing a significant exposure risk to personnel and contractors. QatarEnergy-IS will control exposure to hazardous substances utilizing NIOSH Guidelines to determine the hazard and recommended TLV's for those substances listed in the latest edition of the NIOSH Pocket Guide to Chemical hazards and QatarEnergy requirements.

4.2 HAZARD RECOGNITION

The recognition of hazardous situations associated with exposure to chemical, physical or biological agents is an essential part of the workplace hygiene protection effort. This aspect of the program requires that all personnel with responsibility for the IHP shall be knowledgeable about the health/hygiene hazards of the chemical/physical/biologic agents to which those visiting or working at QatarEnergy-IS facilities may be exposed to. In addition, personnel involved in occupational health/hygiene hazard assessments in QatarEnergy-IS operations shall be knowledgeable regarding tasks personnel conduct under normal, upset, shutdown, and start up conditions, and of the associated risks to health/hygiene. In this context, knowledge of processes and/or equipment is vital.

The IHPP shall be enhanced by the assignment of personnel, knowledgeable in the process, the responsibility to participate, with HSE personnel, in identifying health/hygiene hazards and evaluating exposures. These same personnel shall help to reduce the risk to an acceptable level through implementation of better work practices, by elimination of problems, by improvement of engineering controls and by use of appropriate PPE.

4.3 HAZARD EVALUATION

When necessary, assessments of recognized hazardous chemical/ physical/ biological agents shall be carried out to determine their potential risk to personnel on a case-by-case basis. There are two assessment approaches used to evaluate potential health/hygiene hazards: Qualitative Exposure Assessments and Quantitative Exposure Assessments. The Qualitative Exposure Assessment approach involves identifying exposure factors and then subjectively (qualitatively) evaluating the exposure, while the Quantitative Exposure Assessment) approach objectively measures the exposure to the worker.

4.4 HAZARD CONTROL

The most effective method of industrial hygiene hazard control is to anticipate the hazard and eliminate it before an exposure can occur. For example, the application of sound attenuating material or muffler in the design and construction of a compressor facility is often an effective approach to eliminate noise problems before they occur. Installing these controls eliminates the need for employee training, exposure monitoring, hearing testing, providing hearing protection to employees and ensuring its use, as well as a number of administrative issues. The engineering approach, however, is not always possible and it is necessary to consider alternatives.

The three accepted methods of Industrial Hygiene Hazard Control are through engineering means, administrative methods and the use of PPE. The selected method shall be the most effective and practical based on properties of the chemical/physical agent, environmental considerations, regulatory issues and cost.

4.4.1 Engineering Controls

Engineering controls consist of items which either eliminate an exposure hazard, or which eliminates the need for personnel to be exposed to the hazard (i.e., remotely operated equipment for noisy areas). This type of controls is the primary approach used by QatarEnergy-IS to reduce personnel exposure to Industrial Hygiene hazards. As part of the design review process within the PRM program, personnel exposure is evaluated to determine if extra protection can be implemented in the design phase to reduce the need for further administrative and personal control measures.

4.4.2 Administrative Controls/Work Practices

If engineered solutions to workplace exposures are not practicable or not possible, QatarEnergy-IS uses as a second exposure control method, administrative controls and work Procedures. This type of control is meant to utilize information about the exposure and to put in place practices which either reduce the exposure time or exposure dose by such methods as reducing area personnel, conduct high heat related operations during night hours, and requiring routine break schedules or the posting of continuous monitors to assure that exposures are not occurring. In the course of their activities, all personnel shall be encouraged to look for areas where work practices can be improved to reduce exposures.

Administrative Procedures also encompass employee hygiene education. QatarEnergy-IS employs personnel and contractors from many different countries. These people have varied backgrounds and understanding regarding public health/hygiene, sanitation, and occupational health/hygiene issues. Personnel working regularly offshore or onshore have significant potential to be exposed to hazardous chemicals and physical agents that can cause adverse health/hygiene effects. Those personnel working in workshops, storage areas or other areas/Departments have occasional/limited/selective exposure potentials.

Personnel working in operations shall be knowledgeable with respect to personal sanitation and public health & hygiene practices as well as the workplace health & hygiene issues. The OHP shall be available as a resource to participate in health/hygiene education presentations to these workers.

4.4.3 Personal Protective Equipment (PPE)

Refer to Procedure for selection and use of PPE (IS-HSE-PRC-005).

4.5 FOLLOW-UP AND ROUTINE ASSESSMENT TO ENSURE PROGRAM EFFECTIVENESS

The area Safety Officer, OHP or the IH shall follow-up after needed exposure controls are implemented and as a matter of routine evaluation to ensure that exposures to known Industrial Hygiene hazards remain at or below allowable levels. It may be necessary to monitor the exposure to demonstrate this. The findings of all follow-up and routine assessments shall be documented.

4.6 RECORD KEEPING / STORAGE / RETRIEVAL / ANALYSIS AND REPORTS

Documentation (survey reports, recommendations for reducing an exposure risk, follow-up study information, program audit results, and other information) related to the activities of HSE personnel carrying out their role in the implementation and maintenance of the IHPP shall be maintained in the HSE Department's files.

4.7 QUALITATIVE EXPOSURE ASSESSMENT (TO BE CONDUCTED BY QatarEnergy-IS PERSONNEL)

4.7.1 General

Qualitative Exposure Assessment involves the following steps to complete its cycle. The HSE Manager assumes a lead role towards this.

4.7.2 Identifying Potential Exposure hazards

The first step in controlling hazardous occupational exposures shall be qualitatively assessing the exposure of the chemical/physical/biological agents to which the individual(s) is (are) exposed by evaluating such factors as:

- Name of the agent
- The likelihood for the worker to be exposed to the agent
- Duration of the exposure
- Frequency of the exposure
- Route of the exposure (i.e., for chemical agents, whether skin contact, inhalation, or both)
- Toxicity of the chemical agent
- Physical properties such as temperature, vapor pressure etc. of the chemical agent
- Exposure controls (Administrative, Engineering and/or PPE) currently being used.

A qualitative exposure assessment shall conclude whether additional exposure controls (e.g., engineering, administrative or use of PPE) are necessary to protect worker health/hygiene. HSE Department personnel shall continue to identify hazardous exposure situations and interact with the various Departments to reduce risks to hygiene where necessary.

4.7.3 Prioritizing Exposure Hazards

Information developed in a qualitative exposure assessment will be useful in prioritizing which work situation poses the greatest industrial hygiene risk, and which shall be addressed first to reduce the risk. Consideration shall be given to the time required to implement a permanent fix of the problem. For example, consider a noisy pump that is leaking condensate vapours from the pump seal. Reducing the leak would be the priority. However, this may require shutdown of the pump. A decision to use respiratory protective equipment and hearing protection would be a reasonable approach as an interim control.

4.7.4 Recommending Exposure Controls

The most desirable method for controlling occupational exposures to hygiene hazards is through engineering methods. However, process limitations, cost factors and practical considerations may preclude this approach. In such instances, alternatives including administrative methods and PPE are considered.

HSE Department personnel shall work with the applicable Department to affect the most practical control method for the hazard. The fact that exposure controls are necessary to reduce an exposure risk shall be Documented, along with information on the control method to be used to reduce the risk.

4.7.5 Identifying Exposures to be Quantified

A potential outcome of a qualitative assessment is that a quantitative exposure assessment shall be carried out to determine if the current exposure has exceeded the TLV.

4.8 QUANTITATIVE EXPOSURE ASSESSMENT - FOR SCOPING INDUSTRIAL HYGIENE SURVEYS.

4.8.1 General

It is sometimes necessary to measure the concentration/level and duration of exposure to a chemical/physical/biological agent in order to determine whether an individual's exposure is excessive, based on established and accepted occupational exposure limits. Measurement shall be accomplished employing established monitoring methods (NIOSH or equivalent).

When the result of such a quantitative assessment exceeds the established limit, then the selection and implementation of exposure controls is necessary. Trained individuals shall carry out Quantitative Exposure Assessments.

Industrial Hygienist shall conduct quantitative exposure assessments when deemed necessary to determine worker exposure to chemical or physical agents. These quantitative exposure assessments shall be conducted over a period to better define the exposure profile by taking into account operating and employee variability. They shall be performed during normal operating conditions, during maintenance and shutdown activities and/or when exposures have been historically found to be elevated. Findings shall be Documented and communicated to the Management.

4.8.2 Identifying Assessment Method

Before a Quantitative Exposure Assessment can be initiated for determining an exposure to a specific airborne contaminant or physical agent, the necessary calibrated sampling/monitoring equipment shall be available. In addition, for airborne contaminants, it will be necessary to have the appropriate sampling media (filter/absorption tube/liquid), to know the proper sampling rate, the total sample volume to collect the storage/shipping requirements and the expiration date of the media. It is inadvisable to collect any air samples for quantifying an exposure until a laboratory has been contacted to ensure they can analyse the sample employing a validated method. In other words, sampling should be done as per AIHA or equivalent Standard/approved Procedures and by qualified/experienced personnel. The sampling methodology must also take into account the analytical methods to be used, the sensitivity and detection limits.

4.8.3 Obtaining Necessary Resources

The Industrial Hygienist shall determine the resources needed to enable quantification of exposures.

4.8.4 Executing the Assessment

The determination of a worker's exposure to a hygiene hazard shall be carried out with the cooperation of the worker and the worker's supervisor. Information relevant to the exposure shall be Documented. This shall include the worker’s name, job classification, work area/location, tasks performed, start and stop time of the assessment, PPE worn, as well as other items necessary to comprehensively Document the work environment conditions at that time.

It may be appropriate at times to monitor not only for excursions but ceilings, and TWA to ensure that the QatarEnergy-IS acceptable exposure limits are not exceeded. This information is often found with the MSDS.

The monitoring equipment used shall be functionally checked and calibrated immediately prior to and following the carrying out the assessment. The measurement shall be representative of the worker's exposure. Direct measurements of a contaminant concentration shall be made in the breathing zone (near the face of the individual) while the task(s) is being performed. The measurement of a worker's exposure to a physical agent (e.g., noise, ionizing radiation, etc.) shall be made at the location where the individual is working. Personal samples (i.e., placing the sampling device on the person) shall be collected in the breathing zone of the worker.

4.8.5 Reporting Findings

Exposure results shall be communicated in writing to the monitored worker and his supervisor. Posting of the results of industrial hygiene surveys will be done with the name of the workers removed to provide confidentiality. The OHP shall permanently maintain findings.

4.8.6 Recommending Exposure Controls

The most desirable method for controlling exposure to occupational health & hygiene hazards is the use of engineering methods. However, process considerations, cost factors and practical issues often prevent using this approach. Alternatives, such as administrative methods or use of PPE are often more appropriate. If an exposure is found excessive relative to established exposure limits or an exposure is due to poor work practice or improper use of PPE, then recommendations shall be made for reducing the risk. The selected control method shall be made based on discussions with the worker's supervisor, Department Manager, and HSE Department. The results of the assessment and the need for exposure control shall be Documented.

4.9 HAZARDOUS CHEMICAL, PHYSICAL AND BIOLOGICAL AGENTS

4.9.1 Hydrogen Sulfide

The Procedure for safe operations and the conduct of personnel in the probable presence of Hydrogen Sulfide (H_2S) is outlined in IS-HSE-PRC-023.

4.9.2 Noise

All employees who are considered to be at risk of noise induced hearing loss are eligible for inclusion to the hearing protection Procedure which is outlined in Procedure for Personal Protective Equipment Selection and use (IS-HSE-PRC-005)

4.9.3 Benzene/Condensate

Benzene, a very toxic compound, is a component of the condensate liquids that accompany the gas extracted from the wells.

Benzene can be absorbed into the body through intact skin, as well as through lung tissue if condensate vapor is inhaled. Exposure to the condensate liquids by skin contact or inhalation of its vapours is to be avoided as it results in exposure to benzene and other organic compounds (e.g., pentanes, hexanes, toluene, xylenes, etc.). Recommended PPE such as gloves, respirators, boots, and others when indicated, based on the job being performed shall be used when there is a potential for exposure to condensate liquids.

Qualitative Exposure Assessments shall be conducted for the various jobs performed on QatarEnergy-IS facilities where exposure to benzene and condensate liquids may occur. Personal monitoring to determine the concentration of benzene exposure measures to control such exposure and use of Quantitative Exposure Assessment shall be implemented if the hazard potential is considered significant.

If personnel are exposed to benzene vapor above the QatarEnergy-IS internal exposure criterion, they shall be referred to the Medic or OHP for surveillance. Trained HSE personnel should conduct the measurement.

Exposure of workers to condensate vapor shall be evaluated concurrent with a benzene exposure assessment. It may be necessary to reduce the condensate vapor exposure even though exposure to benzene is acceptable. This decision shall be based on the analytical findings of the exposure assessment

4.9.4 Ionizing Radiation

QatarEnergy-IS has several personnel trained as Radiation Protection Officer (RPO). However, the PS-1 QA/QC Inspector will have the oversight of activities ranging from radiation-related activities done by contractors hired exclusively for the inspection through to measuring and supporting activities dealing with NORM. The RPO shall perform the following:

- Ensure the work complies with IS-FOP-PRC-100 (Integrated Safe System of Work ISSoW) and other approved and applicable Method Statements.

- Ensure that radioactive materials/radiation-producing equipment are received, stored, used, and disposed of by Contractors in a proper and safe manner.
- Audit the activities of contract Radiographers and of radioactive materials/radiation producing equipment to ensure that they do not impact the personnel.
- Use a calibrated Radiation Survey Meter for the measurement of stray radiation levels associated with work or for doing the "Radiation Leakage Test" on the radioactive equipment before allowing its entry into the workplace.
- Maintain records of radioactive materials/radiation-producing equipment in QatarEnergy-IS operations, as well as the radiation exposure records of personnel.
- Review material requisitions for compliance with certification requirements.

The potential for exposure to ionizing radiation exists when radioactive sources are used for Metal/Weld Inspection, operation of interface indicators/instrumentation and Naturally Occurring Radioactive Materials (NORM)**.**

Personnel, who may be exposed to ionizing radiation in their work, shall be provided a personal monitor thermo-luminescent dosimeter (TLD), film badge, and/or pocket dosimeter) to be worn when there is a potential for exposure. Personnel conducting radiography shall cordon off the area, post warning signs/flashing lights around the location where the radioactive source is open, monitor the cordoned periphery during the exposure to ensure no one is exposed to stray radiation. The radiographer shall be equipped with a shielding device for use if the source is not effectively retracted on completion of the exposure and shall use a calibrated survey meter to monitor the area during the Procedure. If demanded on completion of the work, the radiographer shall provide the RPO with a record of the work performed and the radiation levels detected around the work area during the exposure.

The multi-phase flow meters contain a sealed radioactive material, which is to be surveyed and leak tested periodically (e.g., every 6 months). In addition, the shutter mechanism shall be checked on these devices to assure they are operating properly, an area survey carried out, and the labels/tags identifying the source checked during this survey. Since NORM has been identified in process equipment on PS-1 the primary exposure concern is for maintenance personnel working on the equipment. A Manual on NORM, available from the American Petroleum Institute and relevant QatarEnergy procedures, shall be used as guidance to address the occupational and environmental issues of this hazardous material, as well as the recommendations for control, if warranted.

4.9.5 Elevated Temperatures

An outline of heat related illnesses can be found in IS-PRC-HSE-031

4.9.6 Asbestos

Guidelines for the control of asbestos is outlined in IS-PRC-HSE-040

4.9.7 Sulphur Dioxide (SO_2)

Exposure to sulphur dioxide, a respiratory irritant, can occur on the platform as a result of the operation of the flare. Hydrogen sulphide, present in the flared gas, produces sulphur dioxide when burned.

If personnel have to work immediately downwind of a plume containing Sulphur Dioxide, they may need to wear appropriate respiratory protection. The use of an air-purifying full face-piece respirator equipped with an acid gas cartridge will typically provide adequate protection unless the concentration is quite high. (e.g., greater than 20 ppm), which then requires use of a supplied air respirator. If contamination occurs in occupied areas (e.g., control room, living areas, etc.) the ventilation systems should be turned off until the wind direction shifts and the potential for further contamination subsides.

4.9.8 Nitrogen (N_2)

Nitrogen is frequently used during the shutdown as a purging agent for the process equipment and during well work operations. Exposure to nitrogen in confined or semi-confined spaces or near them may prove fatal, if oxygen deficiency in the local atmosphere continues for some time. It affects the nerve-system of a human body and may cause death in absence of adequate oxygen. It does not give enough warning and the victim suddenly faints. Personnel required to enter a confined space which may become oxygen depleted will adhere to the Permit to Work requirements for appropriate respiratory protection.

4.9.9 Lead (Pb)

Exposure to lead can result in serious adverse health/hygiene effects among workers. Excessive exposure to lead may occur if welding, burning, grinding, or cutting is carried out on surfaces containing lead-based paint. Measures shall be implemented to ascertain whether any paint-system used in the plants is lead-based. If so, personnel performing the previously indicated activities shall be required to wear disposable coveralls and an air-purifying respirator appropriate for exposure to lead when welding, burning, grinding or cutting on painted surfaces.

4.9.10 Cadmium (Cd)

Cadmium plated hardware is often specified in the construction of facilities with operations/emissions that are corrosive to steel. Exposure to cadmium fumes occurs when the coated metal is cut using a torch or when welding on metal with cadmium coating. Cadmium is extremely toxic and is carcinogenic (i.e., causes cancer). Supervisors shall identify cadmium coated surfaces to workers if welding or cutting is to be carried out on these and appropriate respiratory protection (supplied air respirator with full face piece, operated in the pressure demand mode) provided to workers involved in these activities.

4.9.11 Zinc (Zn)

Zinc exposure applies to the welding, burning, and cutting on galvanized (zinc coated) steel surfaces can result in worker exposure to zinc oxide fumes/dust. Exposure to zinc oxide fume/dust can give rise to metal fume fever, a flu-like condition In addition, workers shall be provided respiratory protection (air-purifying respirator equipped with a P100 filter) when there is the potential for exposure to zinc oxide fume/dust.

4.9.12 Welding Gases

The welding gases may contain combination of NOx, CO_2, CO and some inert gases, which may cause suffocation in confined space for the welders unless adequate ventilation measures are adopted.

4.9.13 Biological Agents

Biological agents are the harmful viruses and bacteria that are around us on a daily basis. This occupational hazard may be associated with working in sewerage sumps, diving vessel saturation capsules or any workplace with a multiple employee base. Biological agents find their way through infected person(s) wherever there is a possibility of their transmission through water, food or air. Transmission is primarily due to faecal-mouth cross contamination from poor sanitation and hygiene in the general workplace, dining areas, toilets, showers, offices etc.

In order to combat these agents, QatarEnergy-IS has implemented routine cleaning processes for the offshore living accommodations, and conducts/reviews periodic health and hygiene audits at all QatarEnergy-IS locations and contract vessels. Recommendations for change are supported by the UKOOA Offshore Environment Guideline.

5.0 REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/External Document (if any)	Document Referencing
QatarEnergy standard for HSE Incident Reporting Investigation Learning	QP-HSE-STD-021	Internal Document	Sideward
QatarEnergy procedure for HSE Incident Reporting Investigation Learning	QP-HSE-PRC-022	Internal Document	Sideward
IS Procedures for Personal Protective Equipment Selection & Use	IS-HSE-PRC-005	Internal Document	Sideward
IS Hazard Communication Procedure	IS-HSE-PRC-006	Internal Document	Sideward
IS Health, Safety & Environmental Contractor Management Procedure	IS-HSE-PRC-009	Internal Document	Sideward
IS Respiratory protection procedure	IS-HSE-PRC-013	Internal Document	Sideward
IS Health and Hygiene Inspection procedure	IS-HSE-PRC-019	Internal Document	Sideward
IS Procedure for Hydrogen Sulphide Safety	IS-HSE-PRC-023	Internal Document	Sideward
IS Heat Stress Procedure	IS-HSE-PRC-031	Internal Document	Sideward
IS Office Ergonomics Procedure	IS-HSE-PRC-036	Internal Document	Sideward

IS Integrated Safe System of Work ISSoW	IS-FOP-PRC-100	Internal Document	Sideward
IS Procedure for Handling NORM Contaminated Materials and Equipment	IS-FOPS-PRC-247	Internal Document	Sideward
NIOSH Pocket Guide to Chemical Hazards	CDC	External Document	Sideward

6.0 APPENDIX

APPENDIX A - REVISION HISTORY LOG

REVISION NUMBER: 01 **DOCUMENT REVISION DATE: 08/12/2021**

Item Revised	Revision Description	Page No.
All	Document is updated based on new Management system documentation format and new IS organization structure	-
Remarks:		

APPENDIX B – LIST OF ABBREVIATIONS

Abbreviation	Definition
ACGIH	American Conference of Governmental Industrial Hygienists
AIHA	American Industrial Hygiene Association
IHPP	Industrial Hygiene Protection Program
MSDS	Material Safety Data Sheets
NIOSH	National Institute for occupational Health and Safety
OHN	Occupational Health Nurse
PPE	Personal Protective Equipment
PRM	Process Risk Management
TLV	Threshold Limit values
TWA	Time Weighted Average
UKOOA	United Kingdom Offshore Operators Association